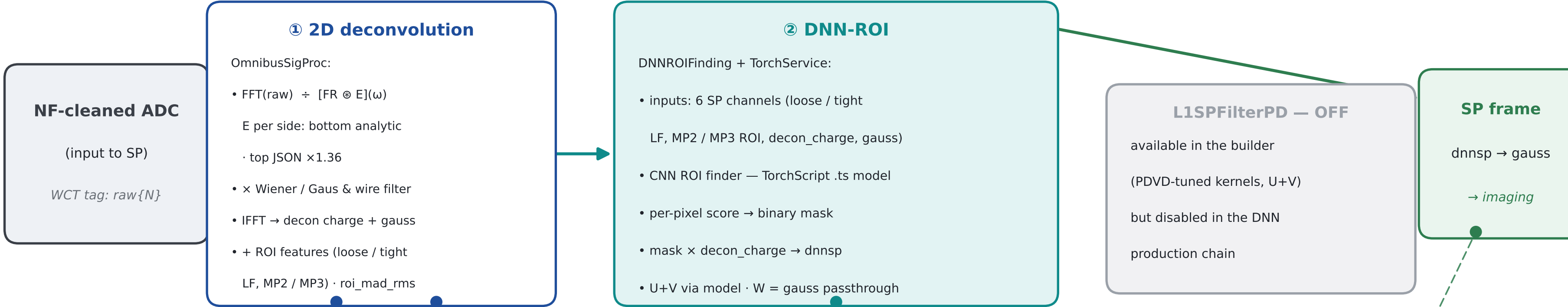
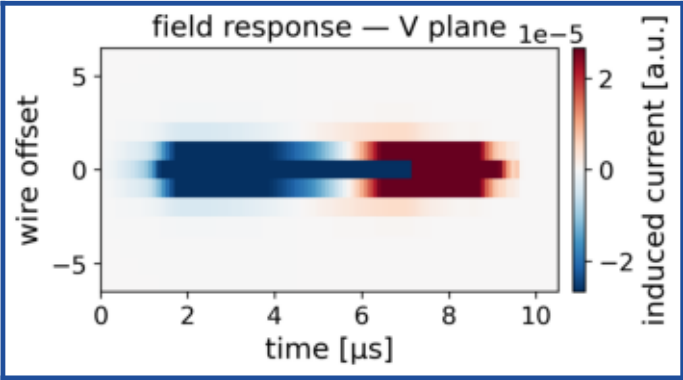


ProtoDUNE-VD Wire-Cell Signal Processing · OmnibusSigProc → DNN-ROI (per CRP)

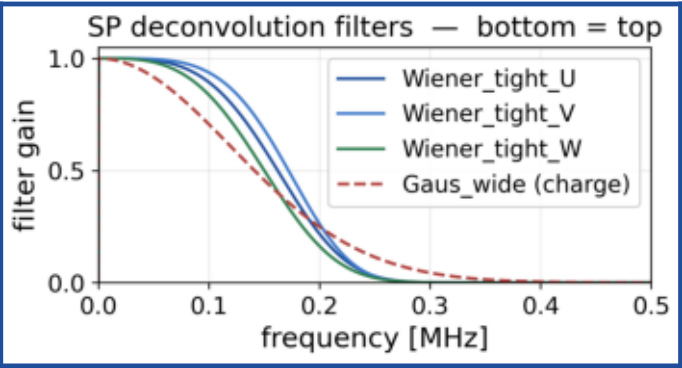
deconvolve $\frac{\text{FFT}[\text{raw}] \cdot G_{\text{filter}}(\omega)}{[\text{FR} * E](\omega)}$ → traditional ROI features → neural-network ROI decision



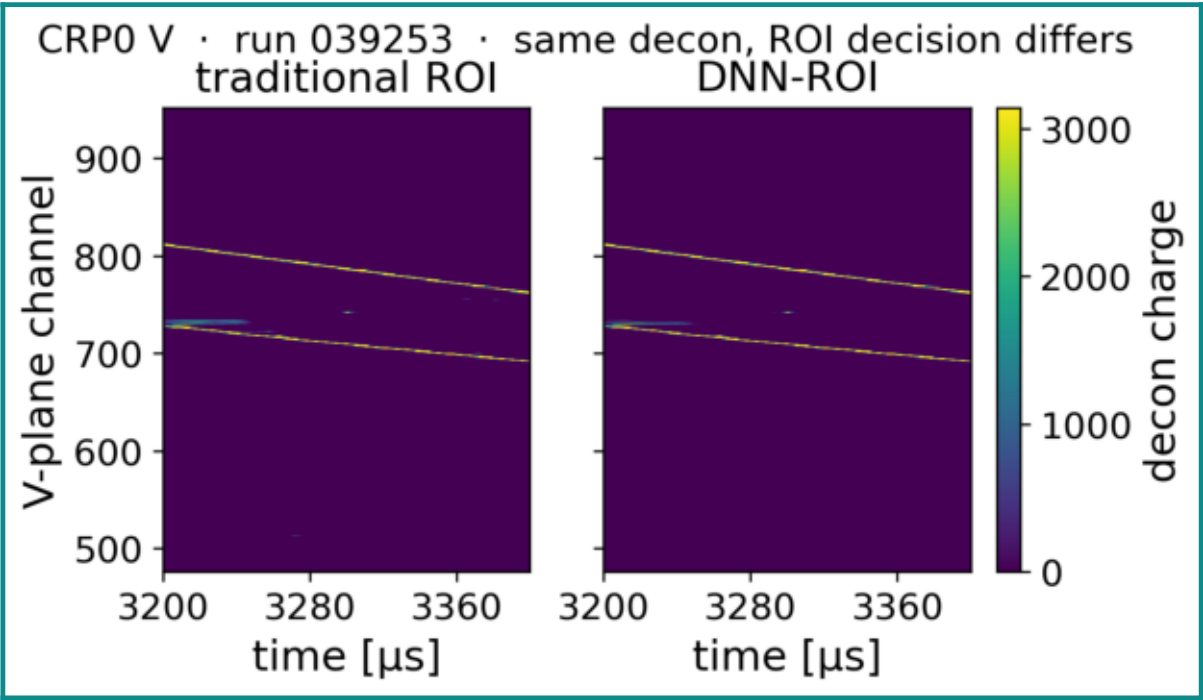
per-side: bottom (idents 0–3) ColdElec 7.8 mV/fC · top (4–7) JSON elec × 1.36, 2.0 V fullscale. W plane: BreakROI off · no DNN (gauss passthrough).



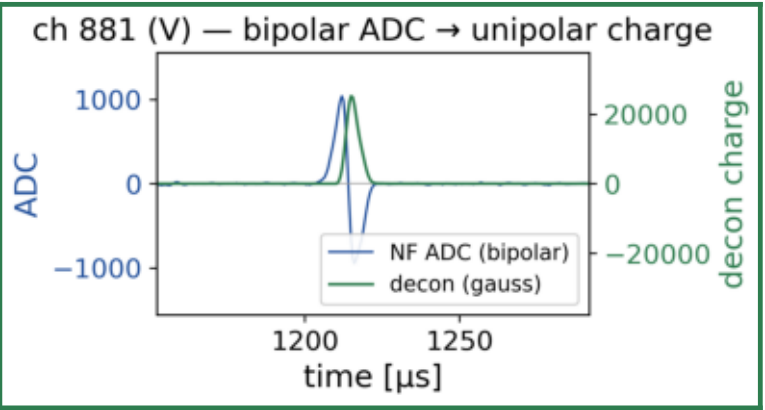
field-response kernel (V)



SP deconvolution filters



traditional ROI vs DNN-ROI (data A/B)



NF ADC → deconvolved charge